

Product Requirements Document (PRD)

House Price Prediction Tool — Bengaluru Real Estate

Prepared for: DUDEX Internship Program — Minor Project (EDA & Basic Prediction)

Version: 1.0 **Date:** July 25, 2026

1. Overview

1.1 Purpose

Build a web-based application that predicts residential property prices in Bengaluru based on location, size, and property attributes, while providing exploratory insights and data-driven investment recommendations to users such as home buyers, sellers, and real estate analysts.

1.2 Problem Statement

Home buyers and sellers in Bengaluru lack a simple, data-driven way to estimate fair property prices across the city's 1,300+ micro-localities. Prices vary significantly by locality, property type, and configuration, and existing listing platforms show asking prices — not predicted fair-market value grounded in data.

1.3 Goals

- Provide an instant, explainable price estimate for a given property configuration.
- Surface locality-level market insights (price trends, undervalued/overvalued areas).
- Demonstrate the full data science lifecycle: cleaning → EDA → feature engineering → modeling → deployment.
- Deliver a portfolio-ready, interactive dashboard suitable for internship evaluation.

1.4 Non-Goals

- Not a real-time listings aggregator (no live scraping of active listings).
 - Not a legal/financial advisory tool — outputs are estimates, not valuations for transactions.
 - Not covering cities beyond Bengaluru in v1.
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2. Target Users

User Persona	Need
Home Buyer	"Is this asking price fair for a 3BHK in Whitefield?"
Home Seller	"What should I list my property at?"
Investor/Analyst	"Which localities are undervalued and likely to appreciate?"
Evaluator (Internship Reviewer)	Assess technical depth, UX, and business insight

3. Scope & Features

3.1 Core Features (MVP — must-have)

F1. Price Prediction Form

- Inputs: location (searchable dropdown), total sq. ft., BHK/bedroom count, number of bathrooms, balconies, area type (Super built-up/Built-up/Plot/Carpet), availability status.
- Output: Predicted price (in ₹ Lakhs), confidence range (e.g., $\pm$ based on model error margin), price per sq. ft.

F2. Model Explainability

- Show top 3–5 factors driving the specific prediction (e.g., via SHAP values).
- Plain-language summary: "This price is driven mainly by location premium and above-average size."

F3. Exploratory Dashboard

- City-wide price distribution (histogram, box plots by BHK).
- Price-per-sq.-ft. heatmap/bar chart by top localities.
- Correlation view: size vs. price, bathrooms vs. price, etc.
- Filters: locality, BHK, price range.

F4. Locality Comparison Tool

- Select 2–3 localities → compare average price/sq.ft., typical configurations, price range.

F5. "What-If" Simulator

- Sliders to adjust sq. ft., bathrooms, BHK → see live price impact without resubmitting the form.

3.2 Enhanced Features (stretch goals — nice-to-have)

F6. Locality Tiering & Investment Insight

- Cluster localities into Premium / Mid / Budget tiers using price statistics.
- Flag "undervalued" localities (below predicted price for their feature profile).

F7. Map View

- Geocode top ~50-100 localities; show price heatmap on an interactive map (Folium/Plotly Mapbox).

F8. Downloadable Report

- Generate a PDF/summary of the prediction + comparable localities for a given query.

F9. Model Version Comparison (technical showcase)

- Toggle between Linear Regression / Random Forest / XGBoost to show performance trade-offs (for evaluator transparency, not typical end-user need).

4. Data Requirements

4.1 Source

- `bengaluru_house_prices.csv` (13,320 rows: area_type, availability, location, size, society, total_sqft, bath, balcony, price).

4.2 Data Preparation Needs

Field	Issue	Treatment
<code>total_sqft</code>	Text, includes ranges ("1200 - 2400")	Parse to numeric, average ranges
<code>size</code>	Inconsistent text ("2 BHK", "4 Bedroom")	Extract integer bedroom count
<code>society</code>	41% missing	Drop or convert to binary flag
<code>location</code>	1 missing; long tail of rare localities	Impute/drop; group rare (<10 listings) into "Other"
<code>bath</code> , <code>balcony</code>	Missing values	Median/mode imputation by BHK group
Outliers	E.g., sqft-per-bedroom implausibly low/high, price anomalies	IQR or domain-rule filtering

4.3 Engineered Features

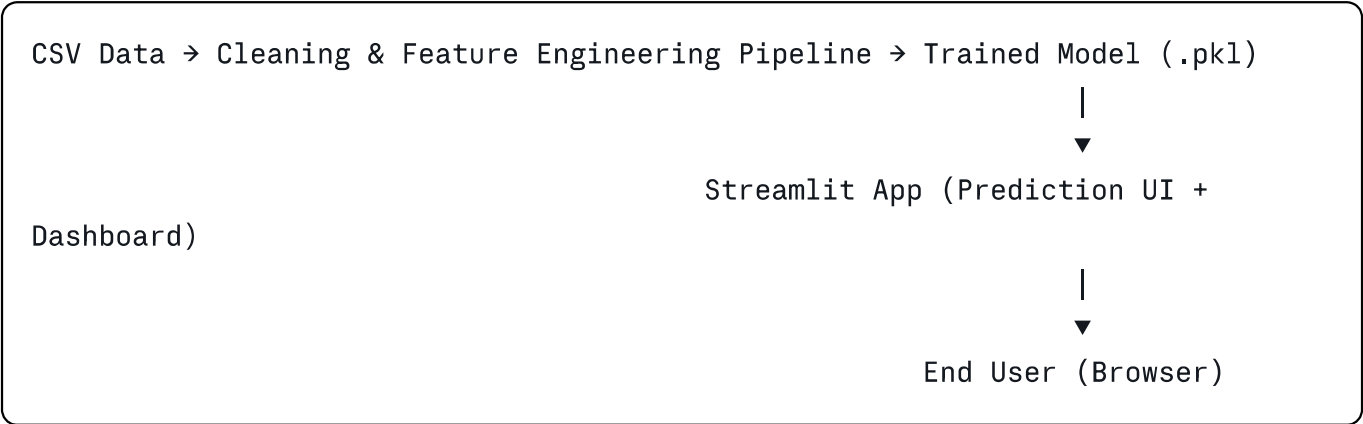
- `price_per_sqft`
- `bhk` (cleaned integer)
- `locality_tier` (clustered)
- `has_society` (binary)
- Optional: distance-to-CBD proxy if coordinates are added later

5. Technical Architecture

5.1 Stack

Layer	Technology
Data processing	Python, pandas, NumPy
Modeling	scikit-learn (Linear Regression, Random Forest), XGBoost/LightGBM
Explainability	SHAP
Web app / frontend	Streamlit (fastest path) <i>or</i> React + Flask/FastAPI backend (if more control over UI is desired)
Visualization	Plotly / Matplotlib / Seaborn (EDA), Plotly for interactive dashboard
Map (stretch)	Folium or Plotly Mapbox
Deployment	Streamlit Community Cloud / Render / local demo

5.2 High-Level Flow



5.3 Model Requirements

- Train/test split with cross-validation.
- Baseline: Linear Regression.

- Primary model: Random Forest or XGBoost (target $R^2 > 0.80$ on test set, informed by typical benchmarks for this dataset).
 - Evaluation metrics logged: RMSE, MAE, R^2 .
 - Model artifact serialized ((`joblib`)/(`pickle`)) and loaded by the app — no retraining at runtime.
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6. UX / Design Requirements

- Clean, single-page or tabbed layout: **Predict** | **Explore** | **Compare** | **About**.
 - Mobile-responsive (Streamlit handles this reasonably by default).
 - Predicted price shown prominently with a confidence range, not a bare number.
 - Visualizations must be interactive where feasible (hover tooltips, zoom).
 - Avoid clutter — progressive disclosure (e.g., "Show factors behind this price" as an expandable section).
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7. Success Metrics

Metric	Target
Model accuracy (R^2)	≥ 0.80 on held-out test data
Prediction latency	< 1 second per request
Dashboard load time	< 3 seconds
Evaluation criteria coverage	All 9 rubric items (creativity, cleaning, EDA, feature engineering, model performance, business insight, dashboard design, innovation, presentation) explicitly demonstrated

8. Business Insights to Deliver (Deliverable, not just app feature)

The final submission should include a short insights write-up covering:

1. Which localities offer the best value (price relative to size/configuration)?
 2. How much does an extra bathroom/balcony actually add to price, holding size constant?
 3. Which property/area types command a premium and why?
 4. Recommendation: 3–5 localities worth watching for investment, with rationale.
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9. Risks & Assumptions

Risk	Mitigation
High missingness in society weakens that feature	Treat as optional binary flag rather than core predictor
Long-tail localities cause noisy predictions	Group rare localities; communicate confidence range
No real coordinates limits true map accuracy	Use approximate geocoding for top localities only, clearly labeled as approximate
Overfitting with too many one-hot location columns (1,305 unique)	Use target encoding or group into tiers instead of raw one-hot

10. Milestones (Suggested Timeline)

Phase	Deliverable	Est. Duration
1. Data Cleaning & EDA	Cleaned dataset + EDA notebook/report	2-3 days
2. Feature Engineering	Final feature set + rationale doc	1-2 days
3. Modeling	Trained models + comparison + chosen final model	2-3 days
4. App Development	Streamlit app (Predict + Explore tabs)	3-4 days
5. Enhancements	What-if simulator, locality comparison, map (stretch)	2-3 days
6. Polish & Presentation	Business insights writeup, README, demo prep	1-2 days

11. Out of Scope for v1

- Multi-city support
- User accounts / saved searches
- Real-time listing integration
- Legal/appraisal-grade valuation guarantees